

1. Find the cartesian equation of the tangent plane to the level surface

$$\mathcal{S} := \{(x, y, z) \in \mathbb{R}^3 : x^2 + \sin(x + y) - 2z^2 = -1\}$$

at the point $(\pi, \frac{\pi}{2}, \frac{\pi}{\sqrt{2}})$.

Set $f(x, y, z) := x^2 + \sin(x + y) - 2z^2 + 1$. The tangent plane to the level surface $\mathcal{S}$ at the point $(\pi, \frac{\pi}{2}, \frac{\pi}{\sqrt{2}})$ has normal vector $\nabla f(\pi, \frac{\pi}{2}, \frac{\pi}{\sqrt{2}})$.

Now,

$$f_x(x, y, z) = 2x + \cos(x + y), \quad f_y(x, y, z) = \cos(x + y), \quad f_z(x, y, z) = -4z. \quad \text{1/2 } \checkmark$$

So,

$$\begin{aligned} \nabla f(\pi, \frac{\pi}{2}, \frac{\pi}{\sqrt{2}}) &= (f_x(\pi, \frac{\pi}{2}, \frac{\pi}{\sqrt{2}}), f_y(\pi, \frac{\pi}{2}, \frac{\pi}{\sqrt{2}}), f_z(\pi, \frac{\pi}{2}, \frac{\pi}{\sqrt{2}})) \\ &= (2\pi, 0, -2\pi\sqrt{2}). \end{aligned}$$

Hence, the tangent plane has cartesian equation

$$\begin{aligned} (2\pi, 0, -2\pi\sqrt{2}) \cdot (x - \pi, y - \frac{\pi}{2}, z - \frac{\pi}{\sqrt{2}}) &= 0 \Leftrightarrow 2\pi(x - \pi) + 0 \times (y - \frac{\pi}{2}) - 2\pi\sqrt{2}(z - \frac{\pi}{\sqrt{2}}) = 0 \\ &\Leftrightarrow \boxed{2\pi x - 2\pi z\sqrt{2} = 0}. \end{aligned}$$

2. Find **all** the critical points of the function $f(x, y) := \sin x + \sin y$ and determine their nature.

We first seek points (x, y) such that $f_x(x, y) = f_y(x, y) = 0$. We find that

$$\begin{cases} \cos x = 0, \\ \cos y = 0, \end{cases} \Leftrightarrow \begin{cases} x = \frac{\pi}{2} + k\pi, & k \in \mathbb{Z} \\ y = \frac{\pi}{2} + 2n\pi, & n \in \mathbb{Z}. \end{cases} \quad (*)$$

So, we find the points

$$\boxed{(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi), \quad k, n \in \mathbb{Z}}.$$

Now, in order to decide whether the points found above are point of maximum, minimum or neither, we need the second order partial derivatives

$$f_{xx}(x, y) = -\sin x, \quad f_{xy}(x, y) = 0, \quad f_{yy}(x, y) = -\sin y.$$

Thus,

$$\begin{aligned} D\left(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi\right) &= \begin{vmatrix} f_{xx}\left(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi\right) & f_{xy}\left(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi\right) \\ f_{xy}\left(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi\right) & f_{yy}\left(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi\right) \end{vmatrix} \\ &= \begin{vmatrix} -(-1)^k & 0 \\ 0 & -(-1)^n \end{vmatrix} = (-1)^{n+k}. \end{aligned}$$

If n is even and k is odd or vice-versa, we find that $n+k$ is odd and hence $D\left(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi\right) = (-1)^{n+k} = -1 < 0$ and hence, the points $\left(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi\right)$ are saddle points.

If both n and k are even, then $n+k$ is even and hence, $D\left(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi\right) = 1 > 0$ and $f_{xx}\left(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi\right) = -1 < 0$. Thus, the points $\left(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi\right)$ are points of local maximum.

If both n and k are odd, then $n+k$ is even and hence, $D\left(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi\right) = 1 > 0$ and $f_{xx}\left(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi\right) = 1 > 0$. Thus, the points $\left(\frac{\pi}{2} + k\pi, \frac{\pi}{2} + n\pi\right)$ are points of local minimum.

3. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) := -x^3 - x \cos(x + y) \quad \text{for every } (x, y) \in \mathbb{R}^2.$$

(a) Find **all** the critical points of the function f .

We seek points (x, y) such that $f_x(x, y) = 0$ and $f_y(x, y) = 0$. We find that

$$\begin{cases} -3x^2 - \cos(x + y) + x \sin(x + y) = 0, \\ x \sin(x + y) = 0. \end{cases} \Leftrightarrow \begin{cases} -3x^2 - \cos(x + y) = 0, \\ x \sin(x + y) = 0. \end{cases} \quad (*)$$

From the second equation, we find that $x = 0$ or $\sin(x + y) = 0$.

For $x = 0$, we get from the second equation that

$$\cos(y) = 0 \quad \Leftrightarrow \quad y = \frac{\pi}{2} + k\pi \quad k \in \mathbb{Z}$$

and we obtain the points $\boxed{\left(0, \frac{\pi}{2} + k\pi\right) \quad k \in \mathbb{Z}}$.

For $\sin(x + y) = 0$, we get

$$\sin(x + y) = 0 \quad \Leftrightarrow \quad x + y = n\pi, \quad n \in \mathbb{Z} \quad \Rightarrow \quad \cos(x + y) = (-1)^n.$$

For $n = 2m$ we get from the first equation in $(*)$ that $-3x^2 - 1 = 0$ which has no solutions.

For $n = 2m + 1$ we get from the first equation in $(*)$ that $-3x^2 + 1 = 0$ which gives $x = \pm \frac{1}{\sqrt{3}}$

and hence, we get from equation $x + y = (2m + 1)\pi$, the points

$$\boxed{\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} + (2m + 1)\pi\right) \quad \text{and} \quad \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} + (2m + 1)\pi\right), \quad m \in \mathbb{Z}}.$$

- (b) Determine the nature of the critical points $(0, \frac{3\pi}{2})$, $(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} + \pi)$, $(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} + \pi)$.

We need

$$\begin{cases} f_{xx}(x, y) = -6x + 2\sin(x + y) + x\cos(x + y), & f_{xy}(x, y) = \sin(x + y) + x\cos(x + y), \\ f_{yy}(x, y) = x\cos(x + y). \end{cases}$$

$$D(0, \frac{3\pi}{2}) = \begin{vmatrix} f_{xx}(0, \frac{3\pi}{2}) & f_{xy}(0, \frac{3\pi}{2}) \\ f_{xy}(0, \frac{3\pi}{2}) & f_{yy}(0, \frac{3\pi}{2}) \end{vmatrix} = \begin{vmatrix} -2 & -1 \\ -1 & 0 \end{vmatrix} = -1 < 0 \Rightarrow (0, \frac{3\pi}{2}) \text{ is a saddle point.}$$

$$D(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} + \pi) = \begin{vmatrix} f_{xx}(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} + \pi) & f_{xy}(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} + \pi) \\ f_{xy}(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} + \pi) & f_{yy}(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} + \pi) \end{vmatrix} = \begin{vmatrix} \frac{7}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{vmatrix} = 2 > 0.$$

So, $D(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} + \pi) = 2 > 0$ and $f_{xx}(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} + \pi) = \frac{7}{\sqrt{3}} > 0$ and hence, $(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} + \pi)$ is a point of local minimum.

$$D(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} + \pi) = \begin{vmatrix} f_{xx}(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} + \pi) & f_{xy}(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} + \pi) \\ f_{xy}(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} + \pi) & f_{yy}(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} + \pi) \end{vmatrix} = \begin{vmatrix} -\frac{7}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \end{vmatrix} = 2 > 0.$$

So, $D(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} + \pi) = 2\sqrt{3} > 0$ and $f_{xx}(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} + \pi) = -\frac{7}{\sqrt{3}} < 0$ and hence, $(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} + \pi)$ is a point of local maximum.

4. Let $(a, b, c) \in \mathbb{R}^3$ be such that $a^2 + b^2 + c^2 \neq 1$. Find on the sphere $x^2 + y^2 + z^2 = 1$, the points which are the closest and the points which are the furthest to the point (a, b, c) .

Hint: It is important to get the right setting, i.e., finding the right function to be minimized and maximized subject to the sphere $x^2 + y^2 + z^2 = 1$.

We want find points (x, y, z) on the sphere which maximize or minimize the function $f(x, y, z) = (x - a)^2 + (y - b)^2 + (z - c)^2$. We set $g(x, y, z) = x^2 + y^2 + z^2 - 1$ and use the method of the Lagrange multiplier for which we seek x, y, z, λ such that

$$\begin{cases} f_x(x, y, z) = \lambda g_x(x, y, z), \\ f_y(x, y, z) = \lambda g_y(x, y, z), \\ f_z(x, y, z) = \lambda g_z(x, y, z), \\ g(x, y, z) = 0. \end{cases} \Leftrightarrow \begin{cases} 2(x - a) = \lambda 2x, \\ 2(y - b) = \lambda 2y, \\ 2(z - c) = \lambda 2z, \\ x^2 + y^2 + z^2 = 1. \end{cases} \Leftrightarrow \begin{cases} (\lambda - 1)x = -a, \\ (\lambda - 1)y = -b, \\ (\lambda - 1)z = -c, \\ x^2 + y^2 + z^2 = 1. \end{cases} \quad (*)$$

In order x, y, z from the first three equations respectively, we need to examine the cases $\lambda = 1$ and $\lambda \neq 1$.

For $\lambda = 1$, we find that necessarily, $a = b = c = 0$ for which the function f is constantly equal to 1 on the sphere $x^2 + y^2 + z^2 = 1$. Thus, each point on the sphere is a point maximum and minimum.

For $\lambda \neq 1$, we find from the first three equations in the system (*) that

$$x = -\frac{a}{\lambda - 1}, \quad y = -\frac{b}{\lambda - 1}, \quad z = -\frac{c}{\lambda - 1}.$$

which substituted in the fourth equation $x^2 + y^2 + z^2 = 1$ give

$$\frac{a^2}{(\lambda - 1)^2} + \frac{b^2}{(\lambda - 1)^2} + \frac{c^2}{(\lambda - 1)^2} = 1 \quad \Leftrightarrow \quad (\lambda - 1)^2 = a^2 + b^2 + c^2 \Leftrightarrow \quad \lambda - 1 = \pm \sqrt{a^2 + b^2 + c^2}.$$

Thus, we get the points

$$\underbrace{A = \left(-\frac{a}{\sqrt{a^2 + b^2 + c^2}}, -\frac{b}{\sqrt{a^2 + b^2 + c^2}}, -\frac{c}{\sqrt{a^2 + b^2 + c^2}} \right)}_{\text{For } \lambda = 1 + \sqrt{a^2 + b^2 + c^2}}$$

and

$$\underbrace{B = \left(\frac{a}{\sqrt{a^2 + b^2 + c^2}}, \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \frac{c}{\sqrt{a^2 + b^2 + c^2}} \right)}_{\text{For } \lambda = 1 - \sqrt{a^2 + b^2 + c^2}}$$

Notice that

$$f(A) = \sqrt{a^2 + b^2 + c^2} + 1 \quad \text{and} \quad f(B) = |\sqrt{a^2 + b^2 + c^2} - 1|.$$

$f(A) > f(B)$ and hence A is the point of maximum while B is the point of minimum.

5. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) := -\frac{1}{4}x^4 + x^2y^2 - 4x^2y \quad \forall (x, y) \in \mathbb{R}^2$.

(a) Find all the critical points of the function f and determine the nature of each point.

We seek points (x, y) such that $f_x(x, y) = 0$ and $f_y(x, y) = 0$. We find that

$$f_x(x, y) = -x^3 + 2xy^2 - 8xy \quad \text{and} \quad f_y(x, y) = 2x^2y - 4x^2.$$

So, we get the systems of equations

$$\begin{cases} -x^3 + 2xy^2 - 8xy = 0, \\ 2x^2y - 4x^2 = 0 \end{cases} \Leftrightarrow \begin{cases} x(-x^2 + 2y^2 - 8y) = 0, \\ 2x^2(y - 2) = 0 \end{cases}$$

It follows from the second equation that $x = 0$ or $y = 2$.

For $x = 0$, the first equation is satisfied for every y and hence all the points on the y -axis are critical points, that is,

$$\boxed{(0, y), \quad y \in \mathbb{R}} \quad (I)$$

For $x \neq 0$, that is, $y = 2$, we obtain

$$\begin{cases} -x^2 - 8 = 0, \\ y = 2 \end{cases} \quad \text{which does not have solutions.}$$

So, the only critical points of the function f are the ones in the box (I) .

Let us now determine the nature of each of those points. To this aim, we need the second order partial derivatives

$$f_{xx}(x, y) = -3x^2 + 2y^2 - 8y, \quad f_{xy}(x, y) = 4xy - 8x, \quad f_{yy}(x, y) = 2x^2.$$

So,

$$D(0, y) = \begin{vmatrix} f_{xx}(0, y) & f_{xy}(0, y) \\ f_{xy}(0, y) & f_{yy}(0, y) \end{vmatrix} = \begin{vmatrix} 2y^2 - 8y & 0 \\ 0 & 0 \end{vmatrix} = 0.$$

Notice that $f(0, y) = 0$ and therefore, to determine the nature of those points, we would like to check whether around those points f is positive or negative. Hence, we should study the sign of $f(x, y)$ around points $(0, y)$.

We find that that

$$f(x, y) = x^2 \left[-\frac{1}{4}x^2 + y^2 - 4y \right] = x^2 \left[-\frac{1}{4}x^2 + (y - 2)^2 - 4 \right]$$

Thus,

$$f(x, y) = 0 \quad \Leftrightarrow \quad x = 0 \text{ or } -\frac{1}{4}x^2 + (y - 2)^2 - 4 = 0.$$

We notice that

$$-\frac{1}{4}x^2 + (y - 2)^2 - 4 = 0 \quad \Leftrightarrow \quad -\frac{x^2}{4^2} + \frac{(y - 2)^2}{2^2} = 1 \quad \left\{ \begin{array}{l} \text{which is a horizontal hyperbola} \\ \text{with } y\text{-intercepts } (0, 4) \text{ and } (0, 0) \\ \text{and asymptotes } y = \pm \frac{x}{2} + 2. \end{array} \right.$$

Therefore,

$$f(x, y) \begin{cases} > 0 & \text{if } -\frac{x^2}{4^2} + \frac{(y - 2)^2}{2^2} > 1 \text{ and } x \neq 0, \\ < 0 & \text{if } -\frac{x^2}{4^2} + \frac{(y - 2)^2}{2^2} < 1 \text{ and } x \neq 0, \\ = 0 & \text{if } -\frac{x^2}{4^2} + \frac{(y - 2)^2}{2^2} = 1 \text{ or } x = 0. \end{cases}$$

Therefore, we can conclude that the critical points $(0, y)$ with $y > 4$ or $y < 0$ are point of local minimum (because we can find a small disc around those points where the function f is positive) and points $(0, y)$ with $0 < y < 4$ are points of local maximum (because we can find a small disc around those points where the function f is negative) while the points $(0, 0)$ and $(0, 4)$ are saddle points (because in any small disc around those points there are points at which the function f is positive as well as points at which the function f is negative).

- (b) Find the points of maximum and minimum of the function f subject to $4x^2 + y^2 - 4y = 0$.

Using the method of the Lagrange multiplier, we seek x, y, λ such that

$$\begin{cases} -x^3 + 2xy^2 - 8xy = 8\lambda x, \\ 2x^2y - 4x^2 = 2\lambda y - 4\lambda, \\ 4x^2 + y^2 - 4y = 0. \end{cases} \quad \Leftrightarrow \quad \begin{cases} x(-x^2 + 2y^2 - 8y - 8\lambda) = 0, \\ 2x^2y - 4x^2 = 2\lambda y - 4\lambda, \\ 4x^2 + y^2 - 4y = 0. \end{cases}$$

From the first equation, we find that $x = 0$ or $-x^2 + 2y^2 - 8y - 8\lambda = 0$.

For $x = 0$ substituted in the last two equation, we find that

$$\begin{cases} 2\lambda y - 4\lambda = 0, \\ y^2 - 4y = 0 \end{cases} \quad \Leftrightarrow \quad \begin{cases} 2\lambda(y - 2) = 0, \\ y(y - 4) = 0 \end{cases}$$

From the second equation, we find that $y = 0$ or $y = 4$ for which we find that $\lambda = 0$ and hence, we find the points $(0, 0)$ and $(0, 4)$.

For $-x^2 + 2y^2 - 8y - 8\lambda = 0$ we get the system

$$\begin{aligned} \begin{cases} -x^2 + 2y^2 - 8y - 8\lambda = 0, \\ 2x^2y - 4x^2 = 2\lambda y - 4\lambda \\ 4x^2 + y^2 - 4y = 0. \end{cases} & \quad (1) \Leftrightarrow \begin{cases} -x^2 + 2y^2 - 8y - 8\lambda = 0, \\ 2x^2(y - 2) = 2\lambda(y - 2) \\ 4x^2 + y^2 - 4y = 0. \end{cases} \\ & \Leftrightarrow \begin{cases} -x^2 + 2y^2 - 8y - 8\lambda = 0, \\ 2(x^2 - \lambda)(y - 2) = 0 \\ 4x^2 + y^2 - 4y = 0. \end{cases} \end{aligned}$$

From the second equation, we find $x^2 = \lambda$ or $y = 2$.

For $y = 2$, we get from the third equation that $x = \pm 1$ and hence, from the first equation that $\lambda = -\frac{9}{8}$. So, we get the points $(-1, 2)$ and $(1, 2)$.

For $x^2 = \lambda$, we find

$$\begin{cases} -9x^2 + 2(y - 2)^2 - 8 = 0, \\ 4x^2 + (y - 2)^2 - 4 = 0. \end{cases} \Leftrightarrow \begin{cases} (y - 2)^2 - 4 = \frac{9}{2}x^2, \\ (y - 2)^2 - 4 = -4x^2, \end{cases} \Rightarrow \frac{9}{2}x^2 = -4x^2 \Rightarrow x = 0$$

which substituted into $(y - 2)^2 - 4 = -4x^2$ gives $y = 0$ or $y = 4$ and once again we get the points $(0, 0)$ and $(0, 4)$ which correspond to $\lambda = 0$.

Now, we evaluate f at the points found above and find that

$$f(0, 0) = f(0, 4) = 0, \quad f(-1, 2) = f(1, 2) = -\frac{17}{4} < 0.$$

Thus, the points $(0, 0)$ and $(0, 4)$ are points of maximum with maximum value equal to 0.

Notice that the ellipse $x^2 + \frac{(y - 2)^2}{2^2} = 1$ (except the points $(0, 0)$ and $(0, 4)$) lies entire in the domain $E := \{(x, y) \in \mathbb{R}^2: -\frac{x^2}{4^2} + \frac{(y - 2)^2}{2^2} < 1\}$ (between the two branches of the hyperbola $-\frac{x^2}{4^2} + \frac{(y - 2)^2}{2^2} = 1$) where the function f is strictly negative.

The points $(-1, 2)$ and $(1, 2)$ are points of minimum with minimum value equal to $-\frac{17}{4}$ and this completes the solution to this question.

6. Find the points of maximum and minimum of the function $f(x, y) = xye^{-xy}$ over the domain

$$D := \{(x, y) \in \mathbb{R}^2: 1 \leq x \leq 4, \quad 0 \leq xy \leq 1\}$$

According to the theory in the lecture notes Maximum-Minimum-IV, we find the points of maximum and minimum in the interior of the domain D and then on its boundary.

- (a) We find the critical points of f lying in the interior of the domain D .

We seek points (x, y) in the interior of the domain D such that $f_x(x, y) = f_y(x, y) = 0$. We find that

$$f_x(x, y) = (y - xy^2)e^{-xy} \quad \text{and} \quad f_y(x, y) = (x - x^2y)e^{-xy}.$$

So, we get

$$\begin{aligned} \begin{cases} (y - xy^2)e^{-xy} = 0, \\ (x - x^2y)e^{-xy} = 0, \end{cases} &\Leftrightarrow \begin{cases} y(1 - xy) = 0, \\ x(1 - xy) = 0, \end{cases} \\ &\Leftrightarrow (x, y) = (0, 0) \text{ or } 1 = xy. \end{aligned}$$

So, we get the points $(0, 0)$ and $(x, 1/x)$ which do not lie in the interior of D .

(b) Now, the boundary of the domain D is made of the following arc of curves:

$$\mathcal{C}_1: \begin{cases} 0 \leq x \leq 4, \\ y = 0; \end{cases} \quad \mathcal{C}_2: \begin{cases} x = 4, \\ 0 \leq y \leq 1/4; \end{cases} \quad \mathcal{C}_3: \begin{cases} 1 \leq x \leq 4, \\ y = \frac{1}{x}; \end{cases} \quad \mathcal{C}_4: \begin{cases} x = 1, \\ 0 \leq y \leq 1; \end{cases}$$

We want to find the points of maximum and minimum for the function f on each of these arcs of curves.

- (i) On $\mathcal{C}_1$ we get the constant function $g_1(x) := f(x, 0) = 0$. So, we get the points $(x, 0)$ with $1 \leq x \leq 4$.
 - (ii) On $\mathcal{C}_2$, we get the function $g_2(y) := f(4, y) = 4ye^{-4y}$ defined on the interval $[0, 1/4]$. We notice that $g_2'(y) = (4 - 16y)e^{-4y} = 4(1 - 4y)e^{-4y} \geq 0$. So, g_2 is an increasing function and hence, achieves its maximum value e^{-1} at $y = 1/4$ and its minimum value 0 at $y = 0$. Thus, we get the points $(4, 0)$ and $(4, 1/4)$.
 - (iii) On $\mathcal{C}_3$, we get the constant function $g_3(x) := f(x, 1/x) = e^{-1}$. So, we get the points $(x, 1/x)$ with $1 \leq x \leq 4$.
 - (iv) On $\mathcal{C}_4$, we get the function $g_4(y) := f(1, y) = ye^{-y}$ defined on the interval $[0, 1]$. We notice that $g_4'(y) = (1 - y)e^{-y} \geq 0$. So, g_4 is an increasing function and hence, achieves its maximum value e^{-1} at $y = 1$ and its minimum value 0 at $y = 0$. Thus, we get the points $(1, 0)$ and $(1, 1)$.
- (c) We obtained in (a) & (b) only the points $(x, 0)$ and $(x, 1/x)$ with $1 \leq x \leq 4$ with $f(x, 0) = 0$ while $f(x, 1/x) = e^{-1}$. Therefore, the points $(x, 0)$ are points of minimum while the points $(x, 1/x)$ are points of maximum.